

Final exam review

Lectures 9–19, in eight pages and sixteen problems

Fall 2026 · Week 12 · the exam is in class, Tue 12/1 — the first day back from recess

How to use this

The exam is Tuesday 12/1, the first day back from Thanksgiving recess. This session is held before recess because there is no session between recess and the exam. Plan accordingly.

The exam covers Lectures 9–19. Read Section 1 first — it is those eleven lectures compressed into the statements you should be able to make without notes — then close it and do Section 2.

The shape of the second half of this course. Lectures 1–8 built machinery. Lectures 9–19 spend it on one repeated question: *does this model tell us anything about the brain, and how would we know?* Almost every lecture presents a method for answering that, then presents a reason the method is not enough. Both halves are examinable, and the second half is where the marks are.

1 Lectures 9–19, compressed

1.1 L09 — Generalization

Training error and test error are different quantities, and only the second one is a measurement of the model. Three splits, three jobs; the test set is opened once.

The **classical bargain**: more capacity lowers training error and eventually raises test error — the U-shaped curve. **Bias–variance** decomposes the error into what the model class cannot express and how much its answer moves with the sample.

Classical theory breaks in three places, and each is examinable. Networks fit **random labels** perfectly (Zhang et al.), so capacity-based bounds cannot explain why they generalize. **Double descent**: past the interpolation threshold, test error falls again. **Grokking**: generalization can appear long after training error reaches zero.

Compositional generalization — systematically recombining known parts — is something humans do easily and networks do poorly. **Shortcut learning**: a model solves the task using a feature that happens to correlate with the label in your data — texture rather than shape, a watermark rather than a diagnosis — and fails out of distribution. This is the recurring failure of the second half of the course.

1.2 L10 — Autoencoders and sparse coding

An autoencoder compresses and reconstructs, and is trained on its own input, so it needs no labels. A **linear autoencoder with tied weights, trained on centered data, recovers the**

PCA subspace — not necessarily the individual components, but the space they span. Add nonlinearity and it can beat PCA, because it can follow a curved manifold.

Sparse coding (Olshausen & Field): reconstruct natural image patches using as few active units as possible. The learned dictionary is oriented, localized, bandpass — Gabor-like, and quantitatively similar to V1 simple-cell receptive fields. This is the course's cleanest example of a *normative* explanation: a computational objective, not a mechanism, predicting the biology.

Efficient coding is the reason to expect this: natural images are redundant, and a code that removes redundancy under a cost constraint is forced toward these filters. **Sparse autoencoders** apply the same idea to a trained network's activations to extract more monosemantic features, which is the current interpretability workhorse.

1.3 L11 — The visual system and CNNs

The ventral stream — retina, LGN, V1, V2, V4, IT — is a hierarchy of increasing receptive-field size, increasing complexity of preferred features, and increasing tolerance to position and size. **Core object recognition** is the behavior to explain: fast, accurate identification despite enormous image variation.

Center-surround in retina and LGN; **orientation-tuned** simple cells in V1, well described by Gabors; complex cells with the same orientation preference and more position tolerance. **HMAX** stacks that: selectivity from template-matching, invariance from max-pooling.

A CNN's convolution is template matching with **weight sharing**; pooling gives tolerance; depth grows receptive fields. Task-optimized CNNs were the first models to predict IT responses well — and models can now be used to *control* neurons via synthesized images, which is a stronger test than prediction.

But: **texture bias** (CNNs classify by texture where humans use shape), brittle out-of-distribution behavior, and a scorecard of human signatures most CNNs do not reproduce.

1.4 L12 — RSA, model-brain comparison, and decoding

Two systems have no shared coordinates, so compare *geometry*. An RDM discards units and keeps pairwise dissimilarity; RSA is the (Spearman) correlation between two RDMs. Interpretation requires a **noise ceiling** and a null obtained by **permuting stimuli**.

The **mapping hypothesis**: a model layer corresponds to a brain area if a (usually linear) map from the model's units predicts held-out responses. Alternatives: RSA, CKA, regression-based scores, and Brain-Score as a standardized aggregate. **The metrics disagree**, high scores can be engineered, and at scale many models score alike — so a single number is weak evidence.

Decoding (MVPA) asks what can be read out of a population. A decoder is a cross-validated classifier; chance depends on the class balance; decodability does not establish that the organism uses the information. An SVM maximizes the margin, and regularization (C) matters most when features outnumber examples, which is the normal case.

1.5 L13 — Explainability

Two families. **Attribution** — saliency (gradient of a logit with respect to the input), Grad-CAM (feature-map level) — is fast and fragile, and can survive randomizing the model's weights, which means it was showing the image rather than the model. **Synthesis** — feature visualization by optimizing an input — requires strong regularization to produce anything interpretable, so the picture is partly a statement about the prior you chose.

Maximally-activating patches are the oldest and often most informative method; **Network Dissection** quantifies it against labeled concepts. Units are frequently **polysemantic**, which motivates sparse-autoencoder features. Meaning generally lives in *directions* in activation space rather than in single neurons. Attribution can catch shortcut learning, which is its clearest scientific use.

1.6 L14–L15 — Recurrence and dynamics

An RNN carries a state: $\mathbf{h}_t = g(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{W}_{xh}\mathbf{x}_t)$, with weights shared across time. Unrolled, it is a deep feedforward network. BPTT differentiates through the unrolled graph, producing a *product* of Jacobians — hence vanishing and exploding gradients. Clipping fixes exploding; **gating** (LSTM, GRU) fixes vanishing, because the cell state passes through an addition.

Viewed as a dynamical system, the same network has **fixed points**. Stable ones are point attractors: discrete memories and decisions. A near-unity eigenvalue gives a **line attractor**: continuous integration and working memory. Trained RNNs are reverse-engineered by finding fixed points numerically and linearizing.

Hopfield networks store patterns via a Hebbian outer product, recall by descending an energy function, and hold about $0.14n$ patterns before spurious blends dominate. Energy-based models generalize this, and connect forward to diffusion. The brain is not feedforward: recurrent circuit models (e.g. the hGRU) solve tasks like Pathfinder where feedforward networks of comparable size fail.

1.7 L16 — Attention and the transformer

Self-attention: $\text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k})\mathbf{V}$, with $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ all linear projections of the same input. Every position sees every other in one step — no long product of Jacobians, and it parallelizes — at the cost of $O(T^2)$ and of losing order, which positional encodings restore. Multi-head is a reshape. A block is attention + MLP with residual connections and Layer-Norm. **ViTs** treat image patches as tokens. Attention maps show routing, not importance.

1.8 L17 — LLMs, scaling, and self-supervision

Next-token prediction, trained with **teacher forcing**. Generation samples from the predicted distribution; temperature and top- k change the output substantially with the weights fixed. **Scaling laws**: loss falls as a power law in parameters, data and compute; Chinchilla showed models were badly under-trained relative to their size. **In-context learning** and **emergent abilities** are claimed; part of the emergence literature is sensitive to the choice of metric, and the exam may ask you to say why.

Self-supervised learning makes the data label itself. Two families: **contrastive** (SimCLR, InfoNCE — augmentations of one image attract, other images repel) and **masked prediction** (BERT, MAE — hide part of the input and predict it). Both learn representations that transfer, without labels, which matters for biology because the brain has no labeled dataset.

1.9 L18 — Generative models

VAE: an encoder outputs a distribution; the reparameterization trick makes sampling differentiable; the ELBO is reconstruction plus a KL to the prior. Latent directions are meaningful and steerable, and VAE latents have been related to IT face-patch coding.

GAN: generator versus discriminator, a minimax game. Sharp samples, unstable training, mode collapse. Latent directions found in a GAN can be used as psychological instruments — GANalyze finds a memorability direction.

Diffusion: add noise with a closed-form forward process, learn to predict the noise, sample by denoising repeatedly. The training objective is a mean squared error; the complexity is in sampling.

1.10 L19 — World models and frontiers

A **world model** is an internal model of environmental dynamics that supports prediction and mental simulation. Whether next-step prediction alone builds a coherent one is open — the taxi-map result suggests it often does not. Evidence for mental simulation in primates; recurrent models learn simulation where feedforward ones do not.

Three limits of current AI: out-of-distribution robustness and shortcut learning; visual and abstract reasoning; data efficiency. And the question the course closes on: *does better AI mean better models of biology?* The trend is worrying — as benchmark performance rises, brain-predictivity has not kept pace — and the proposed diagnosis is to compare *strategies* rather than outputs (ClickMe, harmonization, data diets, better objectives).

2 Problems

1. A network reaches 100% training accuracy on randomly shuffled labels. Why does this undermine capacity-based explanations of generalization?
2. Sketch a double-descent curve. Mark the interpolation threshold and say what is true of the model there.
3. A classifier reaches 94% on a held-out test set and 41% on photographs taken in a different hospital. Name the phenomenon and one method from L13 that could have caught it in advance.
4. Why does a linear autoencoder with tied weights recover the PCA subspace, and why is “the PCA subspace” the right phrasing rather than “the principal components”?
5. Sparse coding on natural images produces Gabor-like filters. Explain in two sentences why this is evidence about V1 rather than a coincidence.

6. What does max-pooling buy, what does it cost, and which cortical cell type is it usually likened to?
7. A model scores 0.42 by RSA against IT. Give the two numbers you need before this means anything, and say how each is obtained.
8. Explain why permuting the entries of an RDM's upper triangle produces a null distribution that is too narrow.
9. You randomize a network's weights and its saliency map is unchanged. What follows?
10. An RNN trains on sequences of length 20 and fails at length 300. Name the mechanism, and say why gradient clipping does not fix it.
11. A trained RNN has a fixed point with one Jacobian eigenvalue at 1.00 and the rest near 0.3. What computation does this support, and give a task that would need it.
12. Why is the $\sqrt{d_k}$ in scaled dot-product attention necessary, and what fails without it?
13. A colleague shows an attention map and claims it explains the model's answer. Give two objections.
14. Teacher forcing makes training faster and creates a problem. State the problem.
15. Explain posterior collapse in a VAE, and give the one-line diagnostic.
16. State the closing question of L19 and the evidence that makes it worrying.

Answers

1. If a network can fit arbitrary labels, its capacity is sufficient to memorize the training set outright — so capacity cannot be what prevents it from memorizing when the labels are real. Whatever explains generalization must therefore involve the data, the optimizer, or the architecture's inductive bias, not the parameter count.
2. Test error falls, rises to a peak at the interpolation threshold, then falls again. At the threshold the model has just enough capacity to fit the training data exactly — roughly, parameters $\approx$ training examples — and the fit is maximally constrained and unstable.
3. Shortcut learning: the model used a feature that correlates with the label in the training hospital (scanner, marking, preprocessing) rather than the pathology. Attribution — a saliency or Grad-CAM map — would show the model attending to a region that could not carry the diagnosis.
4. Minimizing squared reconstruction error through a k -dimensional linear bottleneck is exactly the problem PCA solves, so the optimum spans the top k principal directions. It recovers the *subspace* rather than the components because any rotation within that subspace reconstructs equally well — nothing in the objective orders or aligns the axes.
5. The filters were not fitted to neural data: they fall out of an objective (reconstruct natural images with few active units) applied to natural images alone, and they then match measured V1 receptive fields quantitatively. That makes it a normative explanation — the biology is what you would expect if V1 were solving that problem — rather than a curve fit.

6. It buys tolerance to small shifts (and reduces spatial resolution cheaply); it costs position information, which later layers cannot recover. It is likened to the complex cell, which shares its simple cells' orientation preference but responds across a range of positions.
7. The **noise ceiling**, from a split-half correlation of the neural RDMs, which bounds what any model could achieve; and a **null distribution** from permuting stimulus labels and reindexing rows and columns together, which says what chance looks like.
8. The upper-triangle entries are not exchangeable — they are generated by n stimuli, so moving one stimulus moves an entire row and column together. Permuting entries independently destroys that structure and produces nulls far more disordered than any real relabeling, so the null spread is too small and every observed score looks extreme.
9. The map is determined by the input image and the architecture, not by anything the model learned. Any claim it supported about the trained model's strategy is unsupported; the figure is evidence about edges.
10. Vanishing gradients: BPTT over 300 steps multiplies 300 Jacobians, and the product decays exponentially. Clipping bounds gradients from *above* and does nothing for gradients that are too small; the fix is gating, which routes the cell state through an addition.
11. A line attractor — a direction in state space that neither decays nor grows, with everything else contracting onto it. It supports integrating evidence or holding a continuous value: perceptual decision-making with accumulated evidence, or maintaining eye position or a remembered angle.
12. Dot products of d_k -dimensional vectors grow like $\sqrt{d_k}$, so without the scaling the softmax argument becomes large, the softmax saturates to nearly one-hot, and its gradient nearly vanishes. Training then fails for large d_k while appearing fine for small d_k .
13. (i) There are many heads and many layers, and their maps disagree, so “the” map is a selected one. (ii) Attention weight is not importance — what a position contributes is its value vector, and the residual stream carries information around the attention block regardless.
14. Exposure bias: during training the model always sees ground-truth history, but at generation time it sees its own outputs, including its own errors, which it was never trained to recover from. Errors compound over a sequence.
15. The KL term drives the encoder to match the prior, so $\mu \rightarrow 0$, $\sigma \rightarrow 1$, and the latent code carries no information about the input while a strong decoder reconstructs anyway. Diagnostic: the KL term goes to (near) zero during training.
16. Does better AI mean better models of biology? The worry is that as benchmark performance has risen, brain-predictivity has not risen with it, and at scale many quite different models score alike — so the standard scores no longer discriminate. The proposed response is to compare strategies rather than outputs.

The night before

Eight things worth being able to state cold:

1. the three splits, and what the random-labels result rules out;
2. why sparse coding predicting V1 is a *normative* explanation;
3. selectivity from convolution, tolerance from pooling — and the cortical analogue of each;
4. the noise ceiling and the stimulus permutation, and what each is for;
5. why a saliency map that survives weight randomization is worthless;
6. vanishing gradients as a product, and why gating and not clipping fixes it;
7. what a line attractor is, and one task that needs one;
8. what an attention map does and does not show.